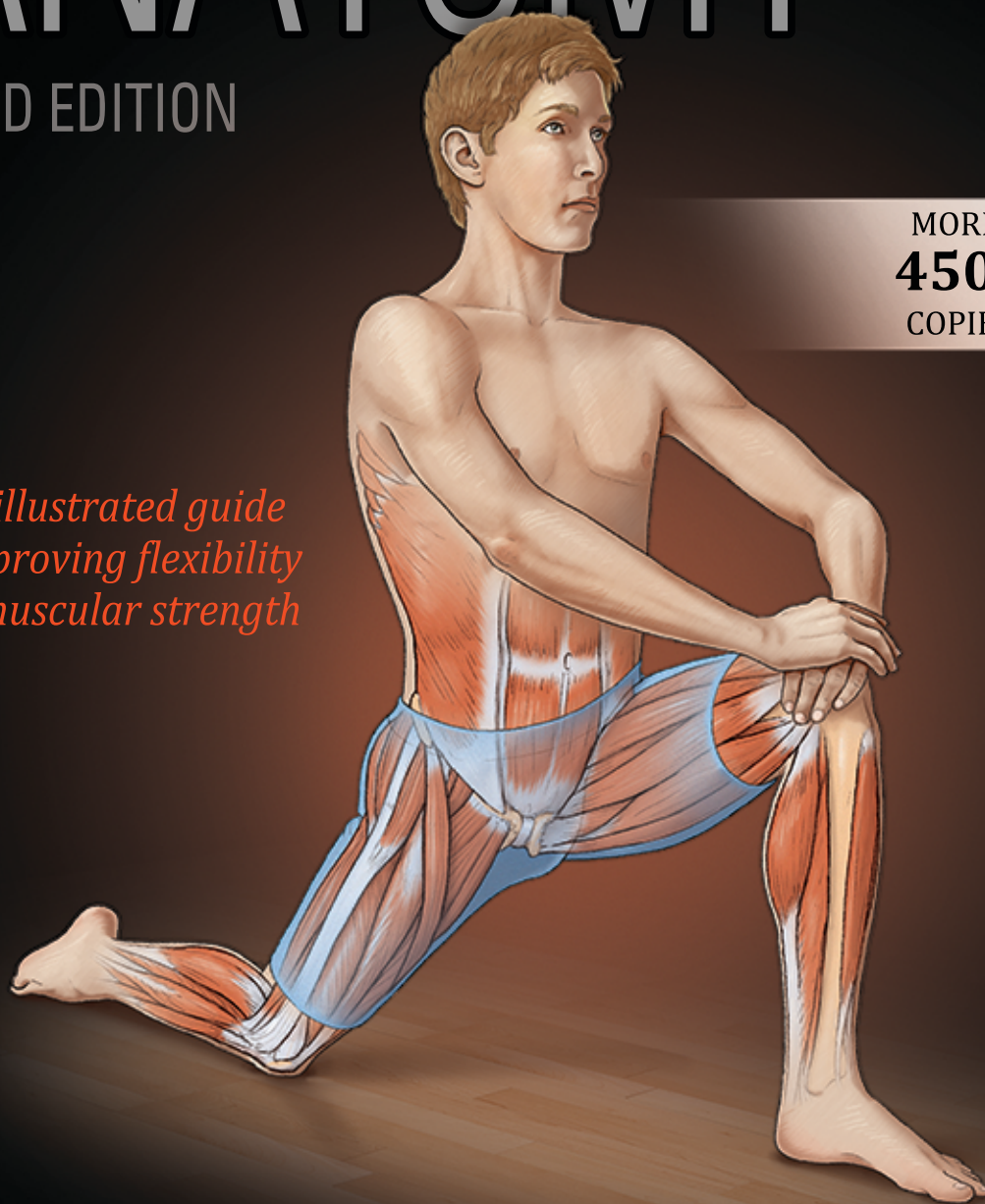


STRETCHING ANATOMY

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ARNOLD G. NELSON | JOUKO KOKKONEN

THIRD EDITION

Stretching

ANATOMY

Arnold G. Nelson
Jouko Kokkonen



HUMAN KINETICS

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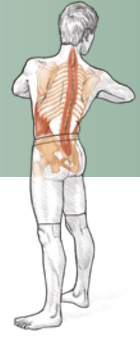
Stretching

ANATOMY

CONTENTS

Preface vi

CHAPTER	1	STRETCHING FUNDAMENTALS	1
CHAPTER	2	FEET AND CALVES	9
CHAPTER	3	KNEES AND THIGHS	37
CHAPTER	4	HIPS	59
CHAPTER	5	LOWER TRUNK	81
CHAPTER	6	ARMS, WRISTS, AND HANDS	107



CHAPTER	7	SHOULDERS, BACK, AND CHEST	141
CHAPTER	8	NECK	173
CHAPTER	9	DYNAMIC STRETCHES	185
CHAPTER	10	PROGRAMS FOR DAILY MOBILITY AND FLEXIBILITY	209
CHAPTER	11	SPORT-SPECIFIC STRETCHING PROGRAMS	219

Stretch Finder 247

About the Authors 251

Earn Continuing Education Credits/Units 254

PREFACE

Although the benefits of regular exercise are well known, few people realize that flexible joints and regular stretching are just as essential for optimal health and activity.

WHY STRETCH?

Flexibility is known to bring positive benefits to the muscles and joints. It aids in injury prevention, helps minimize muscle soreness, and improves efficiency in all physical activities. This is especially true for people whose exercise sessions, whether a recreational game of golf or a more strenuous weekend game of basketball, are more than four days apart. Increasing flexibility can also improve quality of life and functional independence. People whose daily lifestyle consists of long sessions of inactivity such as sitting at a desk can experience a stiffening of the joints so that it is difficult to straighten out of that chronic position. Flexibility helps prevent this by maintaining the elasticity of the muscles and providing a wider range of movements in the joints. It also provides fluidity and ease in body movements and everyday activities. A simple daily task such as bending over and tying your shoes is easier when you are flexible.

Aside from the multiple benefits of stretching for relatively healthy individuals, stretching can offer pain relief and improved mobility to people with chronic conditions such as arthritis. To help relieve pain, especially during the early stages of this condition, people who have arthritis often keep affected joints bent and still. Although holding a joint still and bent may temporarily relieve discomfort, keeping a joint in the same position causes the muscles and ligaments to stiffen. This lack of movement can cause the muscles to shorten and become tight, leading to permanent loss of mobility and a hindering of daily activities. This can lead to less movement, which means fewer calories burned, and added weight puts more strain on the joints. Therefore, fitness experts urge people who have arthritis to stretch all of the major muscle groups daily, placing a gentle emphasis on joints that have decreased range of motion.

Stretching can also help prevent and relieve many muscle cramps, especially leg cramps that occur during the night. The causes of nighttime leg cramps are varied: too much exercise; muscle overuse; standing on a hard surface for a long time; flat feet; sitting for a long time; an awkward leg position during sleep; insufficient potassium, calcium, or other minerals; dehydration; certain medicines such as antipsychotics, birth control pills, diuretics, statins, and steroids; and diabetes or thyroid disease. Regardless

of the cause, a more flexible muscle is less likely to cramp, and stretching helps to immediately reduce the cramp.

Interestingly, current research shows that when stretching exercises are incorporated in a daily or alternate-day program, many problems that accompany the natural aging process can be alleviated. The authors of this book have published several studies in scientific journals showing that stretching routines not only improve flexibility but also increase strength and strength endurance. They have also shown that stretching after weight training is beneficial. In addition, the authors have found that stretching exercises can improve one's ability to maintain balance. For people with impaired mobility, stretching is a viable form of exercise because it has been shown that stretching exercises raise both the heart rate and the whole-body metabolic rate to levels similar to those of a leisurely walk. Finally, in 2011, the *Journal of Physiotherapy* published a research study by Nelson, Kokkonen, and Arnall showing that a program of passive static stretches could lower blood glucose by an average of 18 percent after 20 minutes and 26 percent after 40 minutes. Thus, it is easy to see the benefits of making a stretching program a daily habit.

WHAT'S NEW IN THIS EDITION

The third edition of *Stretching Anatomy* has 10 new stretches to provide even more options for improving flexibility. The four new stretches of the foot and ankle can help to relieve foot problems, especially when sitting for long periods. To help with shoulder problems, three additional static stretches and one new dynamic stretch are available. The chapter on stretches for specific activities has been divided into two chapters. Chapter 10 contains programs to increase flexibility and mobility that will improve performance of activities of daily living and includes stretches for people who sit or stand on their feet all day. Chapter 11 is a new sport-specific chapter and includes stretches for three additional sports.

HOW TO USE THIS BOOK

Chapters 2 through 9 highlight stretches for the major joint areas of the body, beginning with the feet and calves and ending with the neck. Within each chapter are several stretches targeting the muscles involved in moving most of the body's joints. The name of each stretch indicates the major movements of the muscles being stretched. In addition, many of the joint

movements list multiple stretches and are described in order from the easiest to the most difficult. Thus, the movements that are likely to involve the stiffest muscles are presented in a progression. People who are new to a stretching program tend to be less flexible and should begin with the easiest level of stretches. Progression to a more difficult stretch should be made when the participant feels confident that a substantial increase in flexibility has been achieved. This should keep a beginner or person with tight muscles from attempting a stretch that puts too much stress on the joint and could result in muscle, ligament, and tendon damage.

The stretches in chapters 2 through 9 are excellent overall stretches; however, not all of the stretches may be suited to each person's needs. As a rule, to effectively stretch specific muscles, the stretch must involve at least one movement in the opposite direction of the desired muscle's movements. When a muscle has a high level of stiffness, however, use fewer simultaneous opposing movements. As the muscle becomes loose, incorporate more simultaneous opposing movements. It is also recommended that you explore the stretches in this book from different angles of pull. By slightly altering the position of the body parts, such as the hands or trunk, the pull of the muscle changes. This approach is the best way to discover where the tightness and soreness in each specific muscle are located. Exploring different angles while stretching will also bring more versatility to your stretching program. Finally, many of the stretches in chapters 2 through 9 are described for only one side of the body. Similar procedures are to be used for the opposite side of the body.

Chapter 10 suggests stretching programs for daily mobility and flexibility as well as a program shown to lower blood glucose. Finally, chapter 11 includes sport-specific stretching routines. The program tables will guide you to stretches to use in your sport training to ensure that you target the most important muscle groups used in that sport.

The illustrations depict the body positions used for each stretch as well as the muscles being stretched. The muscles most stretched are illustrated in a dark red, and nearby muscles that are less stretched are illustrated in a lighter red. The names of each stretch are based on the actions of the muscles being stretched and not the body and joint positions of the joints. Ideally, to stretch any muscle, the body positions should be opposite to muscle actions (i.e. to stretch a flexor you must extend the joint). A definition of the muscle actions and related terms are presented in the following table.

Most-stretched

Less-stretched

Connective tissues

KEY TERMS

Term	Definition
Abduction	A movement away from the body's midline
Adduction	A movement towards the body's midline
Agonist	Major (prime) mover muscle, works for the desired movement
Antagonist	Acts in opposition to the desired movement
Anterior	The front of the body
Circumduction	A combination of different movements – flexion, extension, abduction, adduction, etc.
Depression	A movement of a body part in a downward direction
Deviation	Flexing the wrist towards one side or the other
Distal	Body parts farther away from the torso of the body
Dorsal	The back (posterior) part of the body and the top of the foot
Elevation	A movement of a body part in an upward direction
Eversion	Turning the sole of the foot in an outward direction
Extension	An increase in the angle of a joint
Flexion	A decrease in the angle of a joint
Inferior	A body part lower than another
Inversion	Turning the sole of the foot in an inward direction
Lateral	Farther away from the midline of the body
Medial	Nearer to the midline of the body
Plantar	Sole or bottom of the foot
Posterior	The back of the body
Pronate	Turn or hold a hand, foot, or limb so that the palm or sole is facing downwards or inwards
Prone	Lying face down
Protraction	A movement of a body part in an anterior (forward) direction
Proximal	Body parts closer to the torso of the body
Retraction	A movement of a body part in a posterior (backward) direction
Superior	A body part higher than another
Supinate	Turn or hold a hand, foot, or limb so that the palm or sole is facing upwards or outwards
Supine	Lying face up

In addition to the illustrations, each stretch contains three sections:

- Execution, which provides step-by-step instructions on how to perform the stretch
- Muscles stretched, which provides the names of the muscles being stretched
- Stretch notes, which provide specific information concerning the how and why behind the need for the stretch as well as safety considerations.